

Power BI Data Model

[dashboards.pbix](#)

Business tables: 12 | Relationships: 10 | Internal auto-date: 6

How to read

1. Next page(s): visual ER diagram (tables + join lines). Zoom in your PDF reader.
2. * = many / FK side (Power BI From). 1 = one / PK side (To).
3. After the ER diagram: every relationship printed in full (nothing clipped).
4. Later pages list every business column (PK/FK marked) and internal date tables.

ER diagram - dashboards.pbix

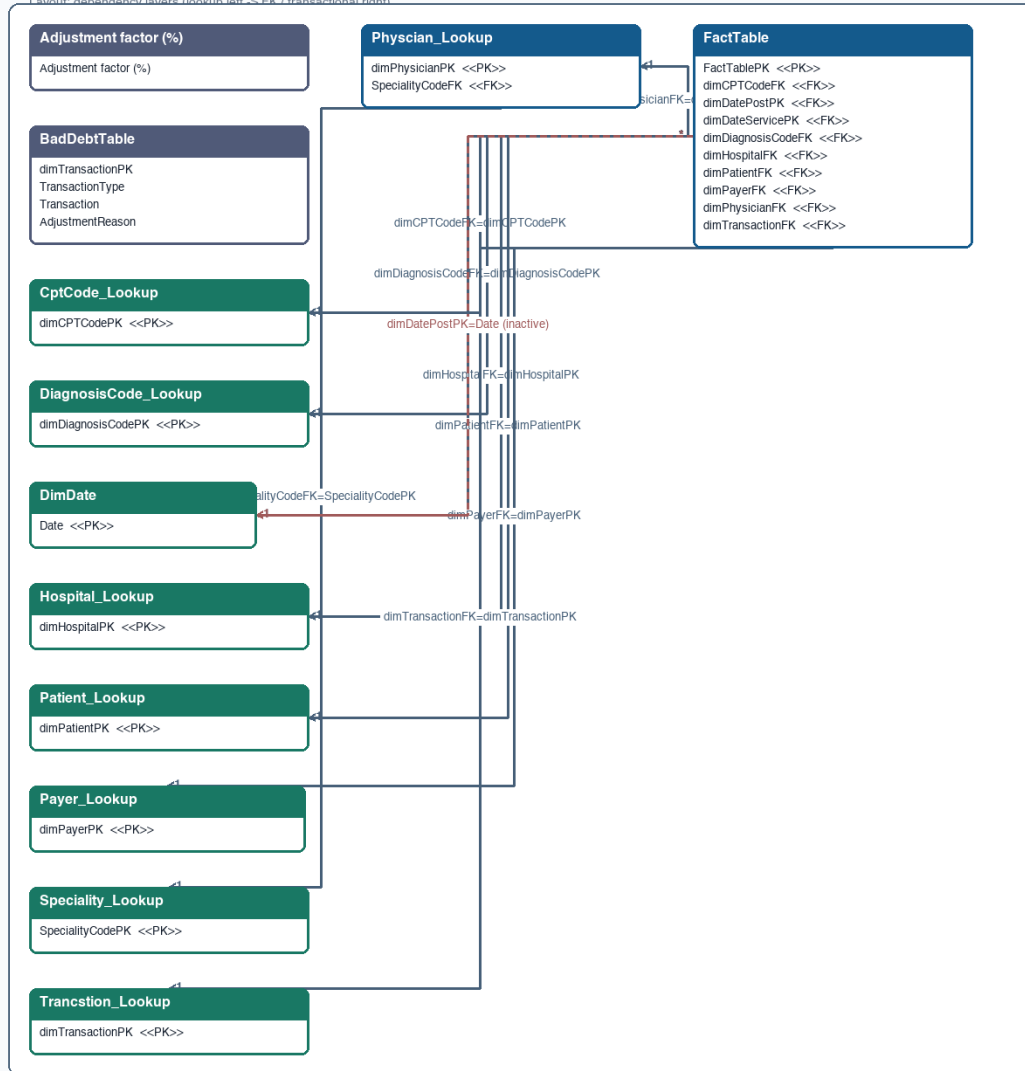
ER diagram

dashboards.pbix

ER diagram (business tables)

Exact joins from the PBIX model. * = many (FK side) 1 = one (lookup / PK side) Dashed red = inactive relationship

Layout: dependency layers (lookup left → FK / transactional right)



ER accuracy: 10/10 model relationships drawn | 12 business entities | From=Many (*), To=One (1)

Entity colors:

■ Fact / FK-heavy

Bridge

■ Dimension / lookup

Other

ER relationships checklist (10)

Many side (FK) -> One side (PK) | Card / Filter / Active

1. FactTable.dimCPTCodeFK -> CptCode_Lookup.dimCPTCodePK [*] many = FactTable [1] one = CptCode_Lookup M:1 filter: Single ACTIVE
2. FactTable.dimDiagnosisCodeFK -> DiagnosisCode_Lookup.dimDiagnosisCodePK [*] many = FactTable [1] one = DiagnosisCode_Lookup M:1 filter: Single ACTIVE
3. FactTable.dimDateServicePK -> DimDate.Date [*] many = FactTable [1] one = DimDate M:1 filter: Single ACTIVE
4. FactTable.dimHospitalFK -> Hospital_Lookup.dimHospitalPK [*] many = FactTable [1] one = Hospital_Lookup M:1 filter: Single ACTIVE
5. FactTable.dimPatientFK -> Patient_Lookup.dimPatientPK [*] many = FactTable [1] one = Patient_Lookup M:1 filter: Single ACTIVE
6. FactTable.dimPayerFK -> Payer_Lookup.dimPayerPK [*] many = FactTable [1] one = Payer_Lookup M:1 filter: Single ACTIVE
7. FactTable.dimPhysicianFK -> Physcian_Lookup.dimPhysicianPK [*] many = FactTable [1] one = Physcian_Lookup M:1 filter: Both ACTIVE
8. FactTable.dimTransactionFK -> Trancstion_Lookup.dimTransactionPK [*] many = FactTable [1] one = Trancstion_Lookup M:1 filter: Single ACTIVE
9. Physcian_Lookup.SpecialityCodeFK -> Speciality_Lookup.SpecialityCodePK [*] many = Physcian_Lookup [1] one = Speciality_Lookup M:1 filter: Single ACTIVE
10. FactTable.dimDatePostPK -> DimDate.Date [*] many = FactTable [1] one = DimDate M:1 filter: Single INACTIVE

Schema layers (left = lookup, right = FK side)

Layer 0: Adjustment factor (%), BadDebtTable (other), CptCode_Lookup (dim), DiagnosisCode_Lookup (dim), DimDate (dim), Hospital_Lookup (dim), Patient_Lookup (dim), Payer_Lookup (dim), Speciality_Lookup (dim), Trancstion_Lookup (dim)

Layer 1: Physcian_Lookup (fact)

Layer 2: FactTable (fact)

Business tables (all columns)

PK / FK marks come from model relationships and key flags.

Adjustment factor (%) (other, 1 cols)

- Adjustment factor (%) (Float64)

BadDebtTable (other, 4 cols)

- dimTransactionPK (Int64)
- TransactionType (string)
- Transaction (string)
- AdjustmentReason (string)

CptCode_Lookup (dim, 4 cols)

- dimCPTCodePK [PK] (Int64)
- CptCode (string)
- CptDesc (string)
- CptGrouping (string)

DiagnosisCode_Lookup (dim, 4 cols)

- dimDiagnosisCodePK [PK] (Int64)
- DiagnosisCode (string)
- DiagnosisCodeDescription (string)
- DiagnosisCodeGroup (string)

DimDate (dim, 20 cols)

- Date [PK] (datetime64[ns])
- Year (Int64)
- Month (string)
- MonthPeriod (Int64)
- MonthYear (datetime64[ns])
- Day (Int64)
- DayName (string)
- Start of Month (datetime64[ns])
- Day of Week (Int64)
- Day Name (string)
- Year.1 (Int64)
- Month.1 (Int64)
- End of Month (datetime64[ns])
- ShortDayName [CALC] (string)
- ShortDayName2 [CALC] (string)
- DayOfWeek [CALC] (Int64)
- Weekend [CALC] (string)
- WeekNum [CALC] (Int64)
- Years [CALC] (Int64)
- Months [CALC] (Int64)

FactTable (fact, 25 cols)

- FactTablePK [PK] (Int64)
- Check Dimension (Int64)
- dimPatientFK [FK] (Int64)
- dimPhysicianFK [FK] (Int64)
- dimDateServicePK [FK] (datetime64[ns])
- dimDatePostPK [FK] (datetime64[ns])
- dimCPTCodeFK [FK] (Int64)
- dimPayerFK [FK] (Int64)
- dimTransactionFK [FK] (Int64)
- dimHospitalFK [FK] (Int64)
- PatientNumber (Int64)
- dimDiagnosisCodeFK [FK] (Int64)
- CPTUnits (Int64)
- Gross Expenses (Float64)
- Adjustment (Int64)
- Insurance_Payment (Int64)
- Patient_Payment (Float64)
- AR (Float64)
- Dayoftheweek [CALC] (Int64)
- CPTUnitType [CALC] (string)
- TotalPayment [CALC] (Float64)
- PatientGender [CALC] (string)
- PatientEthnicity [CALC] (string)
- PatientCity [CALC] (string)
- RegionCode [CALC] (string)

Hospital_Lookup (dim, 3 cols)

- dimHospitalPK [PK] (Int64)
- HospitalName (string)
- IsHospitalNameLong [CALC] (string)

Patient_Lookup (dim, 34 cols)

- dimPatientPK [PK] (Int64)
- PatientNumber (Int64)
- FirstName (string)
- LastName (string)
- Email (string)
- PatientGender (string)
- PatientAge (Int64)
- PatientHeight(in cms) (Int64)
- Year of Birth (Int64)
- Month (Int64)
- Day (Int64)
- BloodGroup (string)

- Tobacco (string)
- Alcohol (string)
- Exercise (string)
- Diet (string)
- Ethnicity (string)
- Zip Codes (Int64)
- State Code (string)
- City (string)
- State (string)
- Region (string)
- DateOfBirth [CALC] (datetime64[ns])
- MonthName [CALC] (string)
- ageofPatient [CALC] (Int64)
- PatientFullName [CALC] (string)
- Age>50 [CALC] (string)
- Patient_AT [CALC] (bool)
- Patient_ATD [CALC] (bool)
- RegionCode [CALC] (string)
- Patient_!AT [CALC] (bool)
- fn [CALC] (string)
- PatientFullName_UPPER [CALC] (string)
- Rounded Age [CALC] (Int64)

Payer_Lookup (dim, 2 cols)

- dimPayerPK [PK] (Int64)
- PayerName (string)

Physician_Lookup (fact, 5 cols)

- dimPhysicianPK [PK] (Int64)
- ProviderNpi (Int64)
- ProviderName (string)
- SpecialityCodeFK [FK] (Int64)
- ProviderFTE (Float64)

Speciality_Lookup (dim, 4 cols)

- SpecialityCodePK [PK] (string)
- ProviderSpecialty (string)
- SpecialityType (string)
- SpecialityDesc (string)

Trancstion_Lookup (dim, 4 cols)

- dimTransactionPK [PK] (Int64)
- TransactionType (string)
- Transaction (string)
- AdjustmentReason (string)

Internal auto-date tables

LocalDateTable_* / DataTableTemplate_* support Power BI date hierarchies. Listed for completeness; business relationships above do not use them.

DataTableTemplate_be97c3f8...

Full name: DataTableTemplate_be97c3f8-f201-4a00-a952-d6592f333bdd

Used for: PBI auto-date TEMPLATE (not tied to a business column)

LocalDateTable_2c2a61ff...

Full name: LocalDateTable_2c2a61ff-39c4-4f0c-97ac-a6a7efb97db4

Used for: DimDate[Start of Month]

LocalDateTable_39bba987...

Full name: LocalDateTable_39bba987-77ed-4008-8322-dd29e2cc2f25

Used for: DimDate[End of Month]

LocalDateTable_a249ec34...

Full name: LocalDateTable_a249ec34-7878-4151-b4c2-6a7b636bf3c0

Used for: DimDate[Date]

LocalDateTable_d31a6fc7...

Full name: LocalDateTable_d31a6fc7-dba3-4410-a53a-de9bb2d5b1c2

Used for: Patient_Lookup[DateOfBirth]

LocalDateTable_ef34a3ff...

Full name: LocalDateTable_ef34a3ff-4085-4eda-a1df-b58a8db8be46

Used for: DimDate[MonthYear]